A Project Report on

CENTRALIZED HACKATHON PORTAL

submitted in partial fulfilment of the requirements for the award of the degree of

BACHELOR OF ENGINEERING

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A.P. SHAH INSTITUTE OF TECHNOLOGY, THANE

2025-2026

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CERTIFICATE

This is to certify that the project entitled "Centralized Hackathon Management Portal" is a bonafide work of **Sumiran Pote (24106113) , Shriya Supekar (24106035) , Raj Surve (24106201) ,Vivek Singh (24106156)** submitted to the University of Mumbai in partial fulfilment of the requirement for the award of the degree of **Bachelor of Engineering** in **Computer Science & Engineering (Artificial Intelligence & Machine Learning**).

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Place:

Declaration

We declare that this written submission represents my ideas in my own words and where others' ideas or words have been included, I have adequately cited and referenced the original sources. I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission. I understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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Date:

Abstract

The rapid growth of technical education has increased the importance of hackathons as a platform for innovation, collaboration, and skill development among students. However, managing hackathons manually or through scattered platforms leads to inefficiencies, lack of transparency, and poor communication.

This project presents a **Centralized Hackathon Management Portal** designed to streamline hackathon-related activities within a college environment. The system provides a unified platform where students can log in to view upcoming and ongoing hackathons, access detailed information, and retrieve problem statements during active events. This promotes the goal of quality education, SDG 4: Quality Education (Promoting skill development and collaborative learning).

A key feature of the portal is the automated progression system, where registered participants are directly entered into the first round and can advance to subsequent rounds based on performance. This reduces manual intervention and ensures a fair and structured selection process.

The portal improves accessibility, enhances user experience, and simplifies hackathon management for both students and organizers. It demonstrates the practical implementation of web technologies, database management, and user authentication systems.

Overall, the system offers an efficient, scalable, and user-friendly solution that modernizes hackathon organization and participation in academic institutions

TABLE OF CONTENTS

1. **Chapter 1** **9**

1.1 Overview **11**

1.2 Need for System: **11**

1.2.1 Scope: 12

2. **Chapter 2** **13**

2.1 Existing System: **14**

3. **Chapter 3** **18**

3.1 Research Gap: **19**

4. **Chapter 4** **20**

4.1. Problem Statement **20**

4.2. Objectives **21**

5. **Chapter 5** **22**

5.1. System Analysis **22**

5.2. Design Details: **25**

6. **Chapter 6** **28**

6.1. Software Requirements **28**

7. **Chapter 7** **30**

7.1. Results: **30**

7.2 Discussion: **31**

7.3 Output: **32**

8. **Chapter 8** **37**

8.1. Conclusion **37**

9. **References** **39**

LIST OF FIGURES

Figure 1; System Architecture of Centralized Hackathon Portal.....24
Figure 2:Home Page 32
Figure 3:Student Login Page 33
Figure 4:Join Portal Page.....33
Figure 5:Student Portal 33
Figure 6:Hackathon Details 34
Figure 7: Registration 34
Figure 8:Profile.....35
Figure 9:Admin Page.....35
Figure 10:Edit Description.....36
Figure 11:Download CSV.....36

LIST OF TABLES

Table 1: Comparison of Existing Systems 16

ABBREVIATIONS:

AI ..... *Artificial Intelligence*

ML ..... *Machine Learning*

UI ..... *User Interface*

UX ..... *User Experience*

DB ..... *Database*

HTML ..... *HyperText Markup Language*

CSS ..... *Cascading Style Sheets*

JS ..... *JavaScript*

API ..... *Application Programming Interface*

SQL ..... *Structured Query Language*

CRUD ..... *Create, Read, Update, Delete*

HTTP ..... *HyperText Transfer Protocol*

HTTPS ..... *HyperText Transfer Protocol Secure*

OS ..... *Operating System*

RAM ..... *Random Access Memory*

IDE ..... *Integrated Development Environment*

URL ..... *Uniform Resource Locator*

JSON ..... *JavaScript Object Notation*

Chapter 1

Introduction

In recent years, hackathons have emerged as a powerful platform for fostering innovation, creativity, and collaborative problem-solving among students. These events provide participants with an opportunity to work in teams, apply their technical knowledge, and develop solutions to real-world challenges within a limited time frame. With the rapid growth of technical education and digital transformation, hackathons have become an integral part of academic institutions, promoting experiential learning and industry-relevant skills.

However, despite their growing importance, the management of hackathons in many colleges remains inefficient and fragmented. Traditional methods rely on multiple disconnected platforms such as emails, messaging applications, spreadsheets, and manual coordination. This scattered approach often leads to lack of transparency, delayed communication, and difficulty in tracking participant progress. Students frequently face challenges in accessing accurate information about upcoming events, retrieving problem statements on time, and understanding their progression across multiple rounds. Similarly, organizers struggle with managing participant data, coordinating event stages, and ensuring smooth execution.

To overcome these limitations, this project proposes the development of a **Centralized Hackathon Portal**, designed to provide a unified and efficient solution for managing hackathon

activities within a college environment. The system acts as a single platform where students and organizers can interact seamlessly. It enables users to access all hackathon-related information in one place, including event details, schedules, rules, and participation status.

One of the key features of the proposed system is the **automated progression mechanism**. Unlike traditional approaches where participant advancement is handled manually, the system automatically promotes users to subsequent rounds based on predefined criteria and performance evaluation. This reduces human intervention, minimizes errors, and ensures fairness and transparency in the selection process.

The portal also enhances accessibility by providing real-time updates and structured information delivery. Participants can easily view ongoing and upcoming hackathons, register for events, and access problem statements during active phases. This eliminates confusion caused by scattered communication channels and improves the overall user experience.

From a technical perspective, the project demonstrates the implementation of modern web technologies, including frontend interfaces, backend processing, database management, and user authentication systems. The system is designed with scalability in mind, allowing future enhancements such as team formation, result analytics, notification systems, and inter-college integration.

Furthermore, this project highlights the importance of digital transformation in academic event management. By replacing manual and fragmented processes with a centralized and automated system, the portal improves efficiency, reduces administrative workload, and enhances user engagement.

In conclusion, the Centralized Hackathon Portal provides a structured, transparent, and user-friendly solution for managing hackathons. It not only improves the experience for students and organizers but also serves as a scalable foundation for future innovations in event management systems.

1.1 Overview

Hackathons are competitive and collaborative events where participants work in teams to solve real-world problems within a limited time frame. These events encourage innovation, critical thinking, and teamwork, making them an essential component of modern technical education.

In a typical hackathon environment, participants are required to register, form teams, access problem statements, and submit solutions within defined rounds. However, managing these activities efficiently requires a structured system that ensures smooth coordination between participants and organizers.

The proposed Centralized Hackathon Portal provides a digital solution to streamline these processes. It offers a unified interface where users can explore hackathons, register for events, track progress, and access important information. By integrating all functionalities into a single platform, the system enhances accessibility, reduces confusion, and improves overall efficiency

1.2 Need for System:

The need for a centralized hackathon management system arises due to several limitations in existing approaches:

- Lack of a unified platform for accessing hackathon information
- Dependence on manual processes for registration and tracking
- Inefficient communication through multiple channels
- Lack of transparency in participant progression
- Difficulty in accessing problem statements during events

These challenges highlight the necessity of a system that can automate processes, centralize information, and provide a seamless experience for both students and organizers.

1.2.1 Scope:

The scope of the proposed system includes:

- Managing hackathons within a college environment
- Providing a centralized platform for students and organizers
- Enabling registration, participation tracking, and result management
- Supporting automated round progression
- Offering scalability for future expansion to inter-college or large-scale events

Chapter 2
Literature Survey/ Existing system

Hackathons have become a significant component of modern technical education, promoting innovation, collaboration, and practical problem-solving skills among students. With the increasing popularity of such events, there has been a growing need for efficient systems to manage hackathon activities. Various platforms and tools have been developed over time to support event organization, participant management, and project evaluation. This section provides an overview of existing systems and approaches used for hackathon and event management. [17]

This work explores how hackathons have become a significant component of modern technical education, promoting innovation, collaboration, and practical problem-solving. It highlights their evolution into multidisciplinary tools that foster critical 21st-century skills. While the research emphasizes the growing popularity of these events, it notes that existing solutions often focus on large-scale industry needs rather than specific academic requirements. Its main contribution lies in identifying the gap between general innovation culture and the lack of integrated pedagogical evaluation metrics in current event tools. [18]

This study provides an overview of the various platforms and tools developed to support event organization, participant management, and project evaluation. It examines how dedicated

platforms offer structured workflows, such as team formation and judging mechanisms, to help organizers manage large-scale events. A major strength identified is the use of real-time dashboards for tracking participant activities. However, the study points out that while effective for global industry hackathons, these systems often lack the customization necessary for smaller, regular academic events. [19]

This research analyzes the limitations of current hackathon management, particularly the reliance on traditional methods like spreadsheets, emails, and disconnected messaging apps. The system analysis reveals that fragmented approaches lead to scattered information and a lack of synchronization, which increases the likelihood of administrative errors. The study advocates for a centralized, automated system that integrates all functionalities—such as round-wise progression and problem statement distribution—into a single platform. While these integrated solutions improve efficiency and user experience, the research notes that high complexity in existing high-end tools remains a barrier for college-level adoption.

2.1 Existing System:

The existing system for managing hackathons in academic institutions typically involves a combination of multiple tools and manual processes rather than a single integrated platform. Students are usually informed about hackathons through emails, social media groups, or notice boards. Registration is conducted using online forms, and the collected data is stored manually in spreadsheets.

During the hackathon, problem statements are often shared through different communication channels such as email or messaging applications. This lack of a centralized distribution mechanism can lead to confusion, delays, and missed updates among participants. Additionally, there is no structured system to manage multiple rounds of the hackathon. Organizers manually track participant performance and decide progression, which increases the likelihood of errors and reduces transparency.

In some cases, external hackathon platforms are used to manage events. While these platforms provide useful features such as dashboards, submission systems, and evaluation tools, they are not specifically designed for college-level requirements. They may lack customization options, require additional setup, or involve subscription costs, making them less suitable for regular academic use.

Another major limitation of the existing system is the absence of real-time updates and centralized access. Students are required to switch between multiple platforms to gather information, which reduces efficiency and negatively impacts user experience. Organizers also face challenges in managing large volumes of data, coordinating different stages of the event, and ensuring smooth execution.

Overall, the current system is fragmented, inefficient, and heavily dependent on manual processes. These shortcomings highlight the need for a centralized and automated solution that can streamline hackathon management and provide a better experience for both students and organizers.

Table 1: **Comparison of Existing Systems**

Sr.No	Research Paper Title	Author(s)	Year	Brief Description / Key Findings
1	Monitoring and evaluation of programming projects in hackathons	*Abduramanov et al.*	2022	Investigates the automation of tools for tracking participant results and checking student work during hackathons to reduce manual error.
2	Hackathons in software engineering education: lessons learned	*Porras, J., et al.*	2018	Analyzes a decade of events to show how centralized platforms can better manage team formation and project evaluation in an academic setting.
3	Driving International Collaboration through Hackathons	*Oyetade, K., et al.*	2024	Discusses the development of an "Open Interactive Digital Ecosystem" (Digital Platform) to facilitate project design and peer-to-peer contact.
4	Effectiveness of Hackathons in Software Engineering Education	*IEEE Frontiers in Education*	2022	Explores how structured platforms help students apply technical foundations (like SOLID principles) by providing a guided project workflow.
5	Hack your organizational innovation: An integrative model	*Heller, B., et al.*	2023	Provides a comprehensive framework for running successful hackathons, highlighting the need for a unified dashboard for organizers and participants.

The above table highlights the comparison between different existing systems used for hackathon management. It can be observed that although each system provides certain useful features, none of them offer a complete and centralized solution tailored for college-level hackathons. This reinforces the need for a dedicated system that integrates all functionalities into a single platform.

Chapter 3
Limitation of Existing system

Despite the availability of various tools and platforms for managing hackathons, the existing systems used in academic institutions suffer from several limitations. These systems are often fragmented, inefficient, and not specifically designed to handle the structured workflow of hackathon events. As a result, both organizers and participants face multiple challenges that affect the overall efficiency and experience of the event. [14]

One of the primary limitations is the lack of a centralized platform. Information related to hackathons, such as announcements, schedules, rules, and problem statements, is distributed across multiple channels including emails, messaging applications, and notice boards. This scattered approach makes it difficult for students to access all relevant information in one place, leading to confusion, missed updates, and reduced participation efficiency. [15]

Another significant issue is the manual registration and data management process. In most cases, registrations are conducted using online forms, and the collected data is stored in

spreadsheets. This method is time-consuming and prone to errors such as data duplication, incorrect entries, and inconsistencies. [16]

Another limitation is the lack of proper tracking and monitoring mechanisms. Organizers often find it difficult to monitor participant progress, evaluate performance efficiently, and maintain records for future reference. The absence of real-time tracking reduces the effectiveness of the event management process and makes it challenging to analyze outcomes.

3.1 Research Gap:

Based on the analysis of existing systems, it is evident that there is a significant gap in providing a **fully integrated and automated solution** for hackathon management in academic institutions. Current approaches rely on multiple independent tools, which results in fragmented workflows, inefficiencies, and poor coordination.

There is no system that combines all essential functionalities such as **event management, participant registration, real-time updates, problem statement distribution, and automated round progression** into a single unified platform. This lack of integration leads to increased manual effort, higher chances of errors, and reduced transparency in the overall process.

Furthermore, existing systems fail to provide a **user-friendly and scalable solution** that can adapt to the specific needs of college-level hackathons. Most platforms are either too complex, not customizable, or not cost-effective for academic use.

Therefore, there is a clear need for a system that can:

- Eliminate dependency on multiple platforms
- Automate key processes such as registration and progression
- Provide centralized access to all hackathon-related information
- Ensure transparency, efficiency, and ease of use

This research gap highlights the importance of developing a **Centralized Hackathon Portal**, which addresses these challenges by offering a unified, automated, and scalable solution tailored specifically for academic institutions.

Chapter 4
Problem Statement and Objectives

4.1. Problem Statement

Current academic hackathons are often hindered by a fragmented approach, relying on disconnected platforms and manual processes that scatter information across emails and messaging apps. This lack of a centralized system makes it difficult for students to track event details, access problem statements, or understand their progression through various rounds. Consequently, the participant experience is marked by confusion and a lack of transparency, as there is no single source of truth for ongoing activities.

For organizers, this manual coordination creates a heavy administrative burden and increases the risk of errors or evaluation bias. Without automated progression and real-time communication tools, maintaining synchronization across event stages becomes nearly impossible. To resolve these inefficiencies, a unified, user-friendly platform is essential to streamline management, integrate all functionalities, and ensure a transparent, professional experience for both students and organizers.

4.2. Objectives

The main objectives of the proposed Centralized Hackathon Portal are as follows:

- To develop a **centralized platform** for managing hackathons within a college environment, ensuring all activities are handled in a single system
- To provide **easy and real-time access** to information about upcoming and ongoing hackathons
- To enable students to **view detailed information** about hackathons, including rules, schedules, and event descriptions
- To allow participants to **access problem statements during active hackathons**, ensuring timely and structured information delivery
- To implement an **automated system for participant progression** across multiple rounds, reducing manual intervention
- To minimize **administrative workload** for organizers by automating registration, tracking, and data management processes
- To improve **user experience and accessibility** through a simple and intuitive interface

Chapter 5
Proposed System

5.1. System Analysis

The proposed Centralized Hackathon Portal is designed to address the limitations of existing systems by providing a unified, automated, and efficient platform for managing hackathon activities. The system eliminates the need for multiple tools by integrating all functionalities such as registration, information access, and participant progression into a single interface.

The system focuses on improving:

- Accessibility of hackathon-related information
- Automation of repetitive processes
- Transparency in participant evaluation
- Efficient data management and tracking

The architecture is designed to ensure smooth interaction between users and the system while maintaining performance and scalability.

Actors in the System

The system primarily consists of two main actors:

- Students(Users):
  Participants who register, view hackathons, access problem statements, and track their progress.

- Organizers(Admin):
  Individuals responsible for creating hackathons, managing participants, updating rounds, and monitoring the overall workflow.

System Framework

The proposed system follows a three-tier architecture, which ensures separation of concerns and improves scalability and maintainability.

1. Presentation Layer (Frontend)

- Provides the user interface for interaction
- Displays hackathon details, dashboards, and results
- Ensures a responsive and user-friendly design

2. Application Layer (Backend)

- Handles business logic and processing
- Manages authentication and authorization
- Controls hackathon workflows and progression

3. Data Layer (Database)

- Stores user information, hackathon details, and participation records
- Ensures efficient data retrieval and storage
- Maintains data consistency and integrity

Workflow Of System

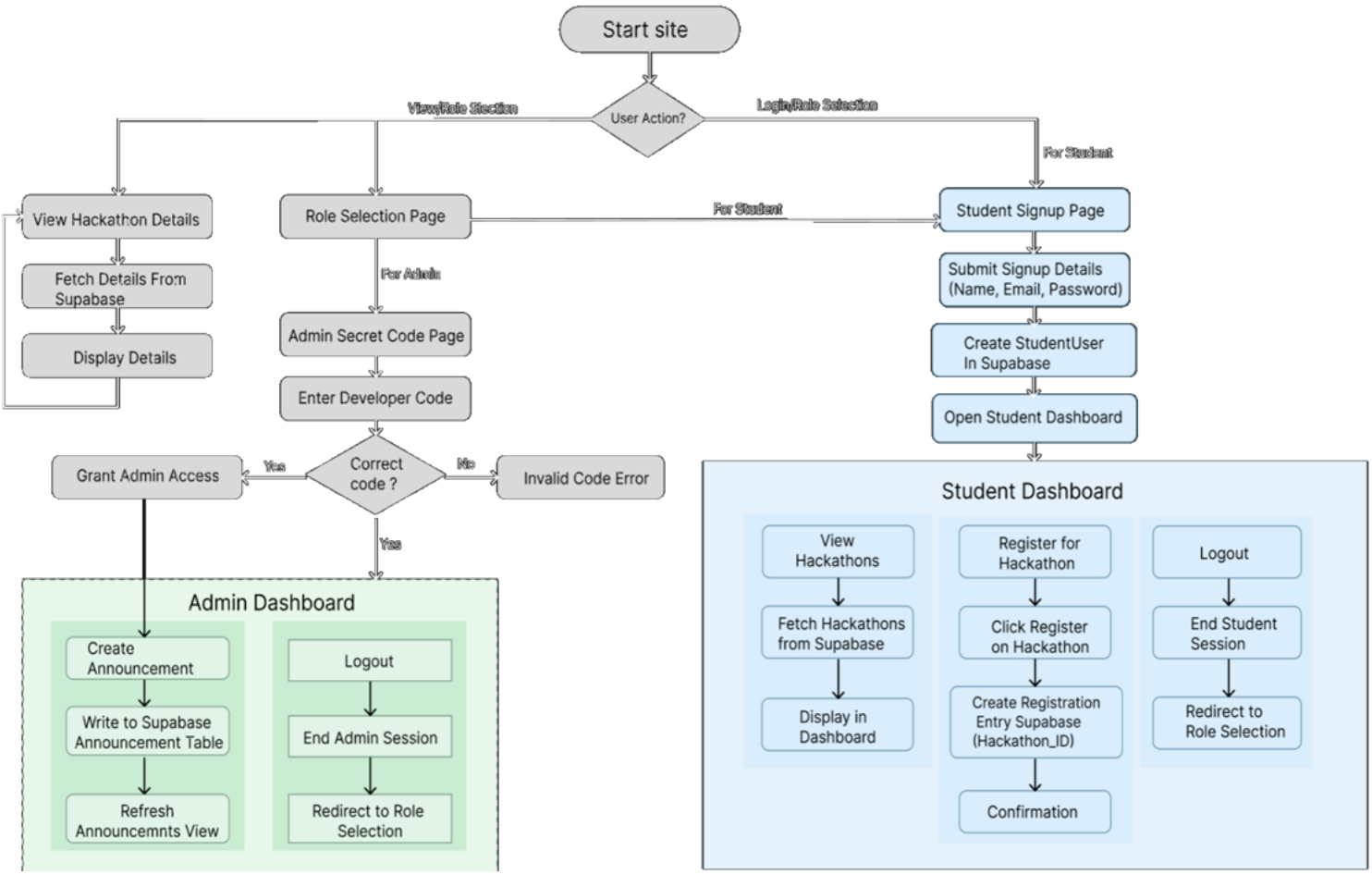

Figure 1; System Architecture of Centralized Hackathon Portal

The working of the system follows a structured and automated sequence of operations that ensures smooth execution of hackathon activities. Initially, the user registers or logs into the system, after which the system securely authenticates the user's credentials to grant access. Once authenticated, the user is able to view a list of upcoming and ongoing hackathons available on the platform. The user can then select a specific hackathon to access detailed information such as rules, schedule, and description. After reviewing the details, the user proceeds to register for the selected hackathon, and upon successful registration, the system automatically enrolls the user into the first round.

During the active phase of the hackathon, the system displays the problem statement to all registered participants in a timely and organized manner. As the event progresses, the system evaluates user performance based on predefined criteria set by the organizers. Based on this evaluation, qualified participants are automatically promoted to subsequent rounds without requiring manual intervention. This process continues until the final stage, where the system generates and displays the results on the portal.

Overall, this workflow ensures a structured, automated, and transparent process for managing hackathons. By minimizing manual effort and streamlining operations, the system enhances efficiency, reduces errors, and improves the overall experience for both participants and organizers.

5.2. Design Details:

The system is designed using a **modular approach**, allowing easy maintenance, scalability, and future enhancements. Each component is developed independently while ensuring seamless integration.

Database Design

The system uses a relational database to store and manage data efficiently. The main tables include:

- **Users Table**
    - User_ID (Primary Key)
    - Name

    - Email
    - Password
- **Hackathon Table**
    - Hackathon_ID (Primary Key)
    - Name
    - Description
    - Date
- **Participation Table**
    - User_ID (Foreign Key)
    - Hackathon_ID (Foreign Key)
    - Round_Status

This structured database design ensures efficient data handling, quick retrieval, and proper relationship management between entities.

Interface Design

The user interface is designed to be **simple, intuitive, and user-friendly**, ensuring ease of use for both students and organizers.

Key features include:

- Dashboard displaying all hackathons
- Easy navigation between pages
- Clear presentation of event details
- Responsive design for multiple devices

The interface minimizes complexity and enhances user interaction with the system.

System Design Considerations

While designing the system, the following factors were taken into account:

- **Security:**
  Implementation of authentication mechanisms and data protection techniques

- **Scalability:**
  Ability to expand the system for larger events or multiple institutions

- **Efficiency:**
  Fast data processing and minimal response time

- **Reliability:**
  Ensuring consistent performance and error handling

Chapter 6
Experimental Setup

The experimental setup of the proposed **Centralized Hackathon Management Portal** includes the configuration of software, hardware, database, and performance parameters required for the successful implementation and execution of the system. This chapter describes the tools, technologies, and environment used to develop and test the system.

6.1. Software Requirements

The system is developed using standard web development technologies. The software requirements are as follows:

- **Operating System:** Windows 10 or above
- **Frontend Technologies:** HTML, CSS, JavaScript
- **Backend Technologies:** (mention yours: PHP / Node.js / etc.)
- **Database:** MySQL
- **Development Tool:** Visual Studio Code
- **Server:** XAMPP / Localhost
- **Web Browser:** Google Chrome

These tools are used to design, develop, and test the web-based application efficiently.

• Operating System: Windows 10 or above
Used as the main platform to run all tools, software, and the project smoothly.

• Frontend Technologies: HTML, CSS, JavaScript
Used to design and make the website look good and interactive for users.

• Backend Technologies: (PHP / Node.js / etc.)
Used to handle logic, process data, and connect frontend with the database.

• Database: MySQL
Used to store and manage project data like user info, records, etc.

• Development Tool: Visual Studio Code
Used to write, edit, and manage code easily with helpful features.

• Server: XAMPP / Localhost
Used to run the project locally and test backend + database without internet.

• Web Browser: Google Chrome
Used to open, test, and debug the website during development.

Chapter 7
Results and Discussion

The proposed **Centralized Hackathon Management Portal** was successfully designed and implemented using web development technologies. The system was tested under different scenarios to evaluate its functionality, performance, and usability. The results demonstrate that the system effectively addresses the limitations of existing hackathon management methods.

7.1. Results:

The following results were observed after implementing the system:

1. User Authentication

- The login and registration system works efficiently
- Users are able to securely access the portal

2. Hackathon Dashboard

- Displays all upcoming and ongoing hackathons clearly
- Information is centralized and easily accessible

3. Hackathon Details

- Users can view complete details including description, schedule, and rules
- Improves clarity and reduces confusion

4. Problem Statement Access

- Problem statements are available during active hackathons
- Ensures timely and organized distribution

5. Automated Round Progression

- Users are automatically entered into Round 1
- System updates participant status based on performance
- Reduces manual work and errors

6. Data Management

- All user and hackathon data is stored efficiently in the database
- Easy retrieval and tracking of participant progress

7.2 Discussion:

The results indicate that the proposed system significantly improves the efficiency of hackathon management compared to traditional methods. The centralized nature of the portal ensures that all information is available in one place, eliminating the need for multiple platforms.

The automated features of the system, such as participant progression and data handling, reduce the workload on organizers and minimize the chances of human error. Additionally, the system provides better transparency, as users can track their participation and progress throughout the hackathon.

The user interface is simple and easy to navigate, which enhances the overall user experience. The system is also scalable, allowing future enhancements such as team formation, notifications, and analytics.

However, the system currently operates within a limited scope and can be further improved by integrating advanced features and expanding its usage beyond a single institution.

7.3 Output:

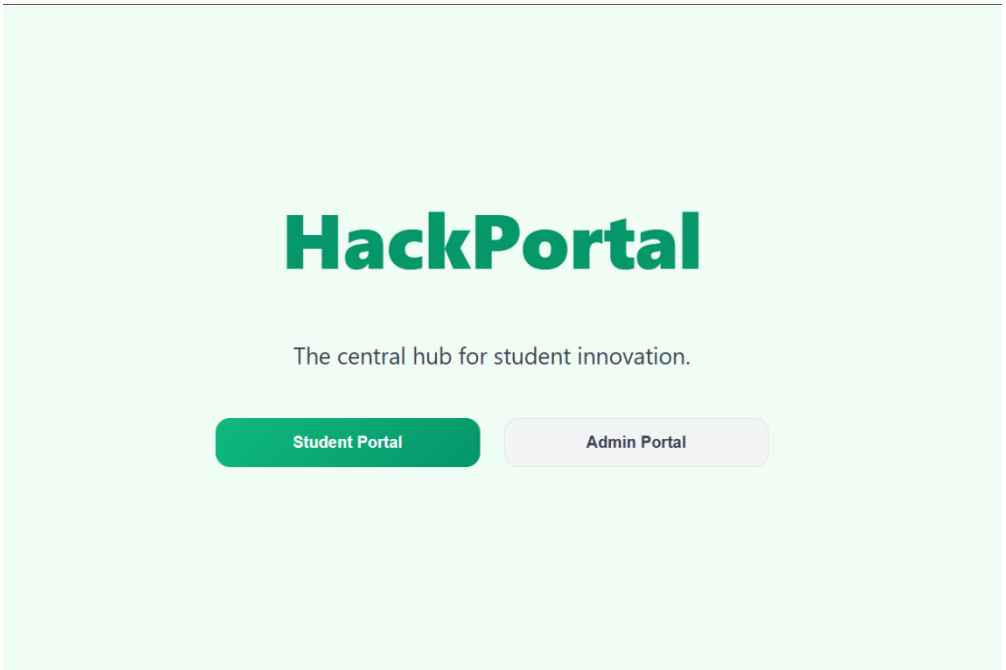

Figure 2:Home Page

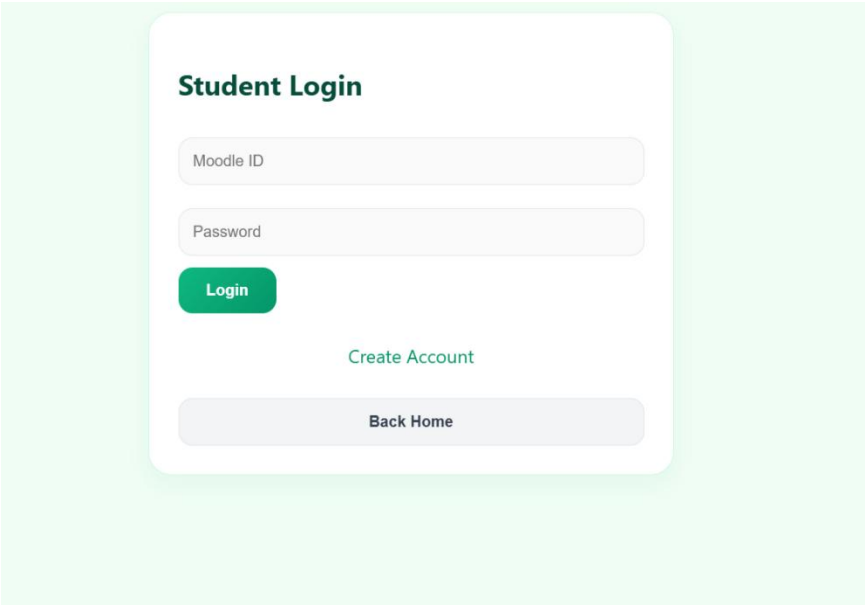

Figure 3:Student Login Page

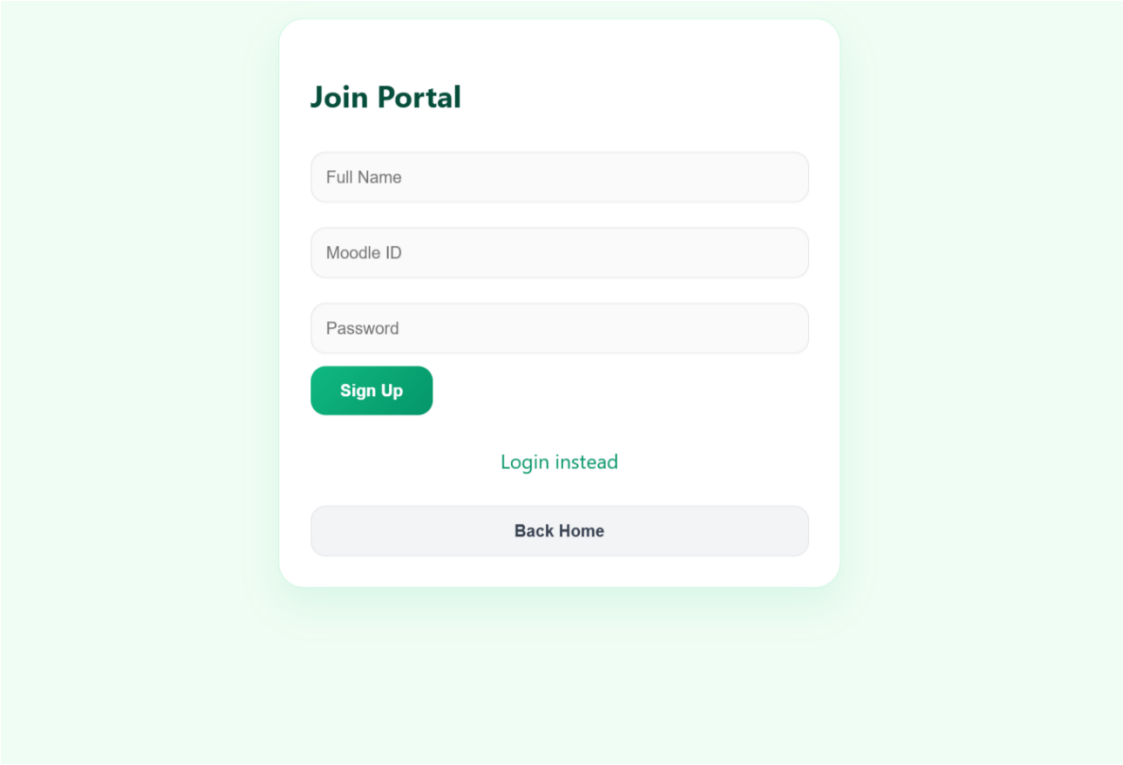

Figure 4:Join Portal Page

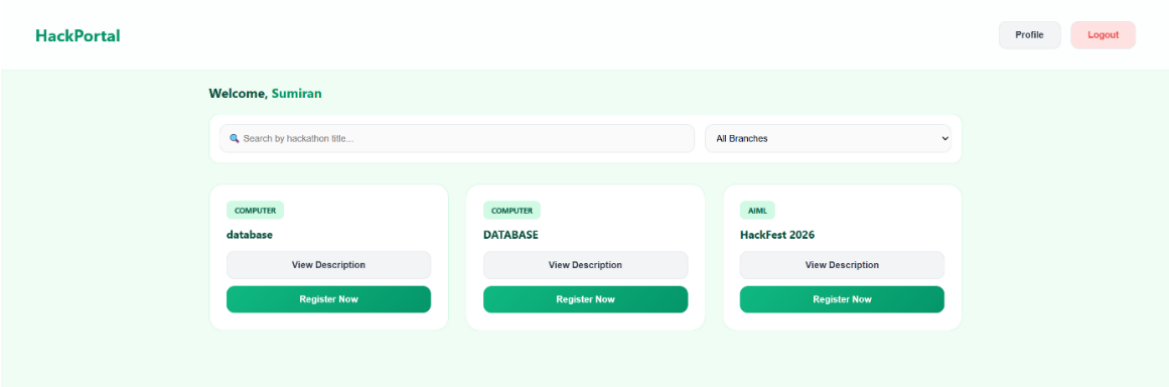

Figure 5:Student Portal

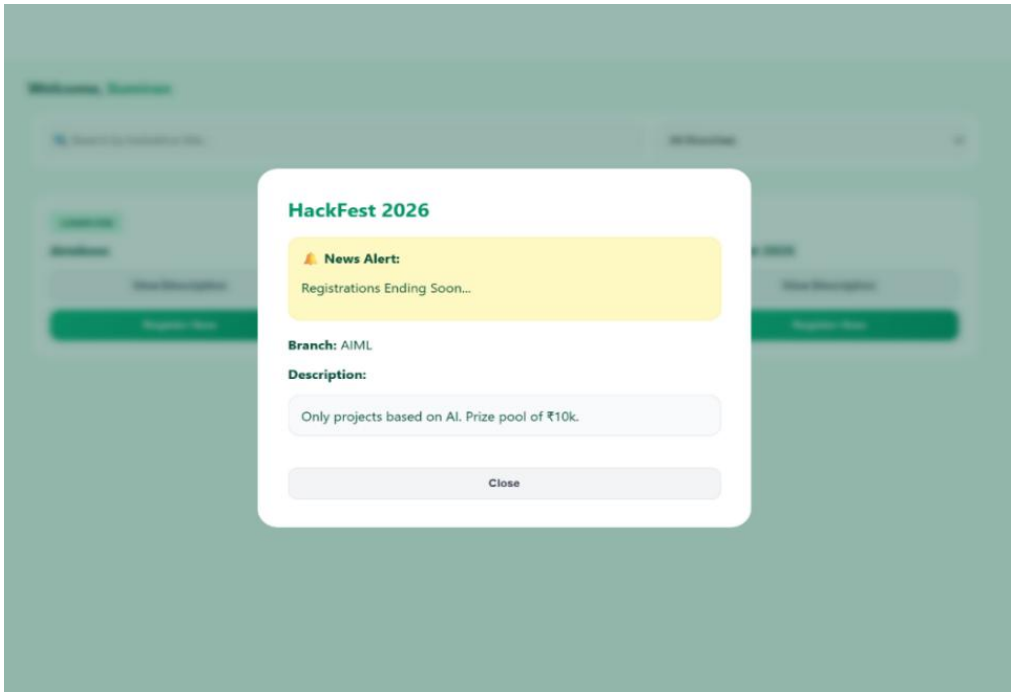

Figure 6:Hackathon Details

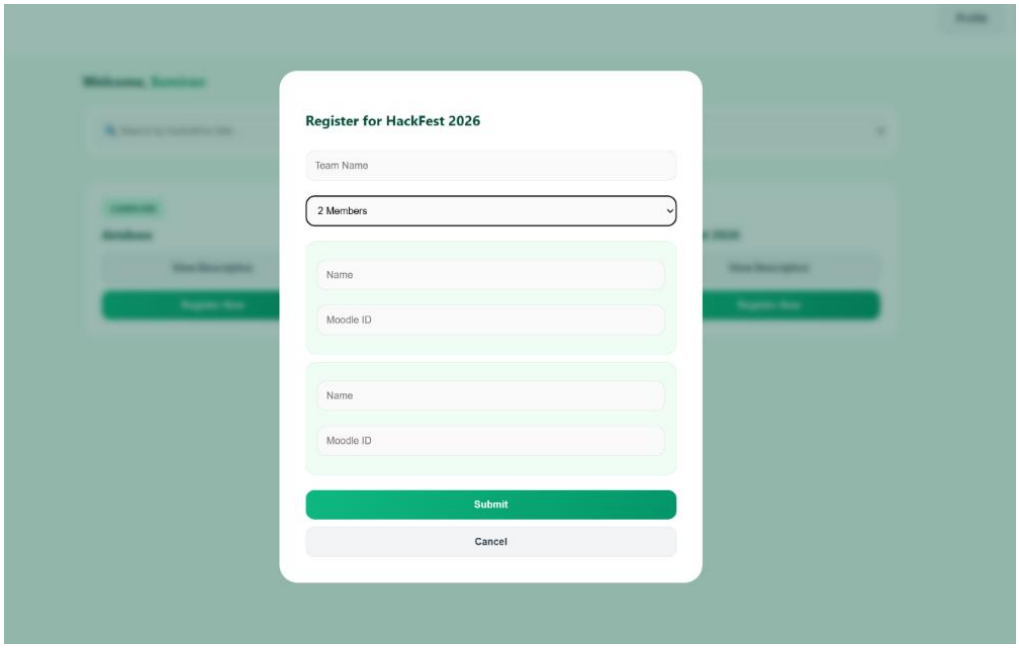

Figure 7: Registration

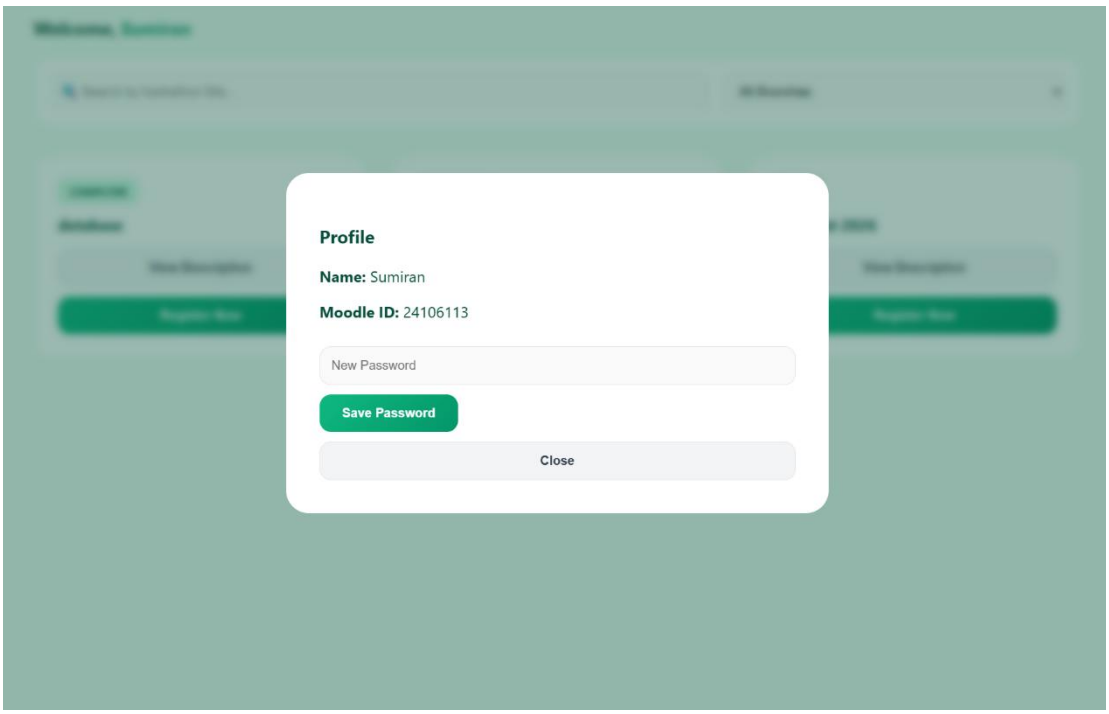

Figure 8:Profile

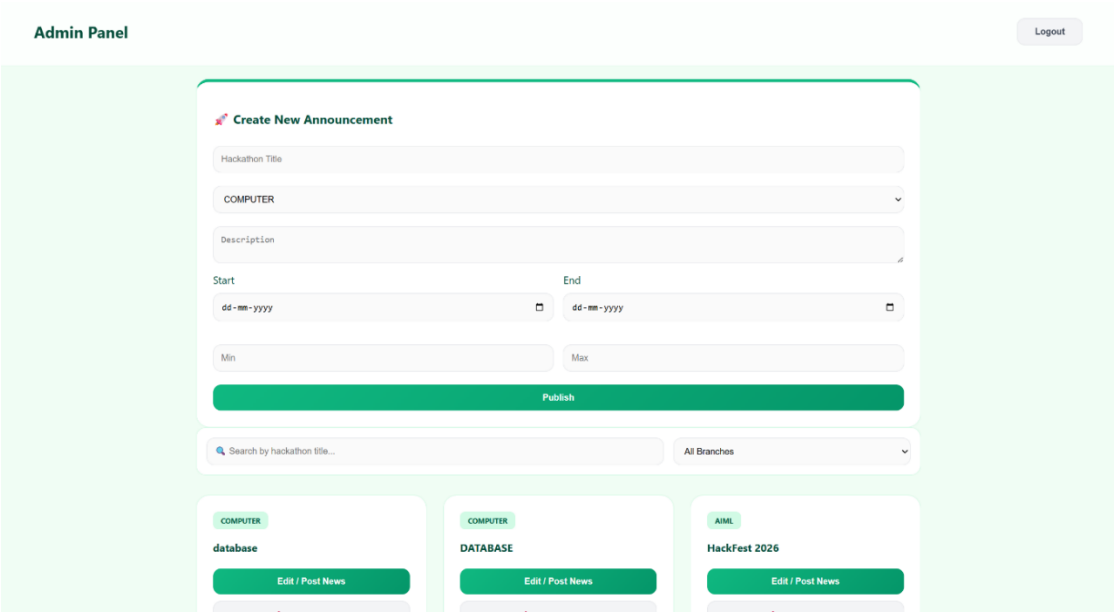

Figure 9:Admin Page

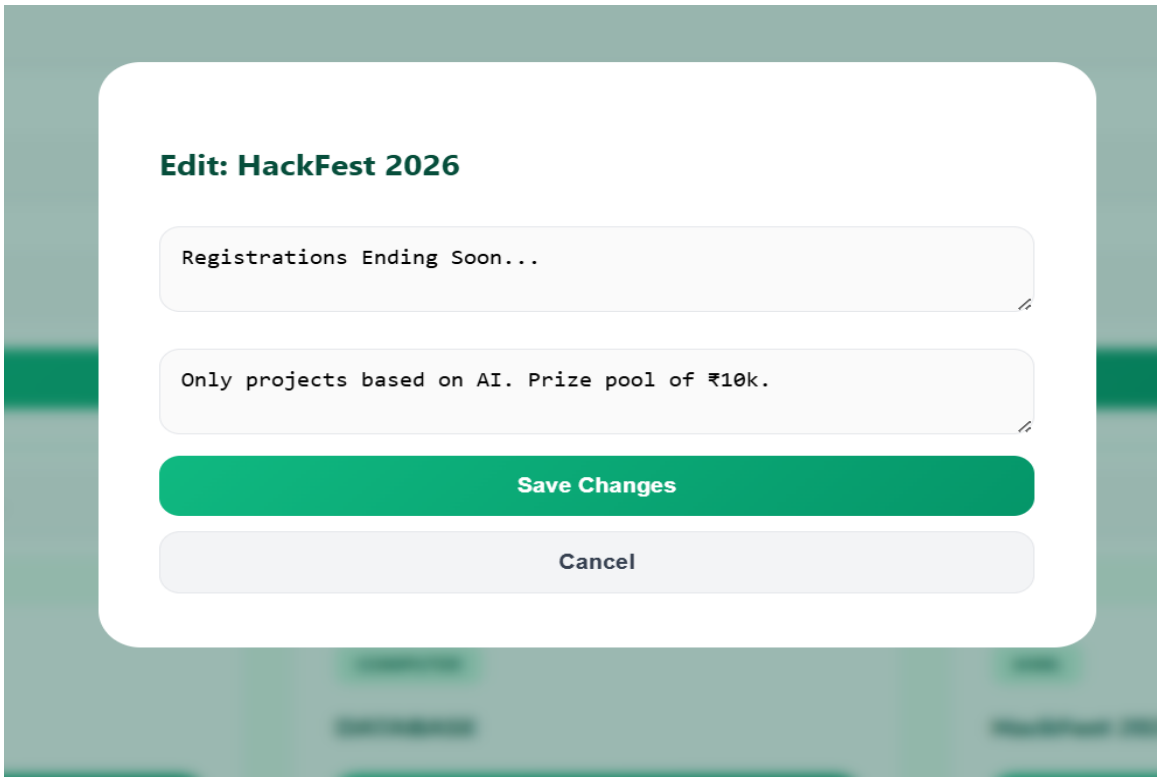

Figure 10:Edit Description

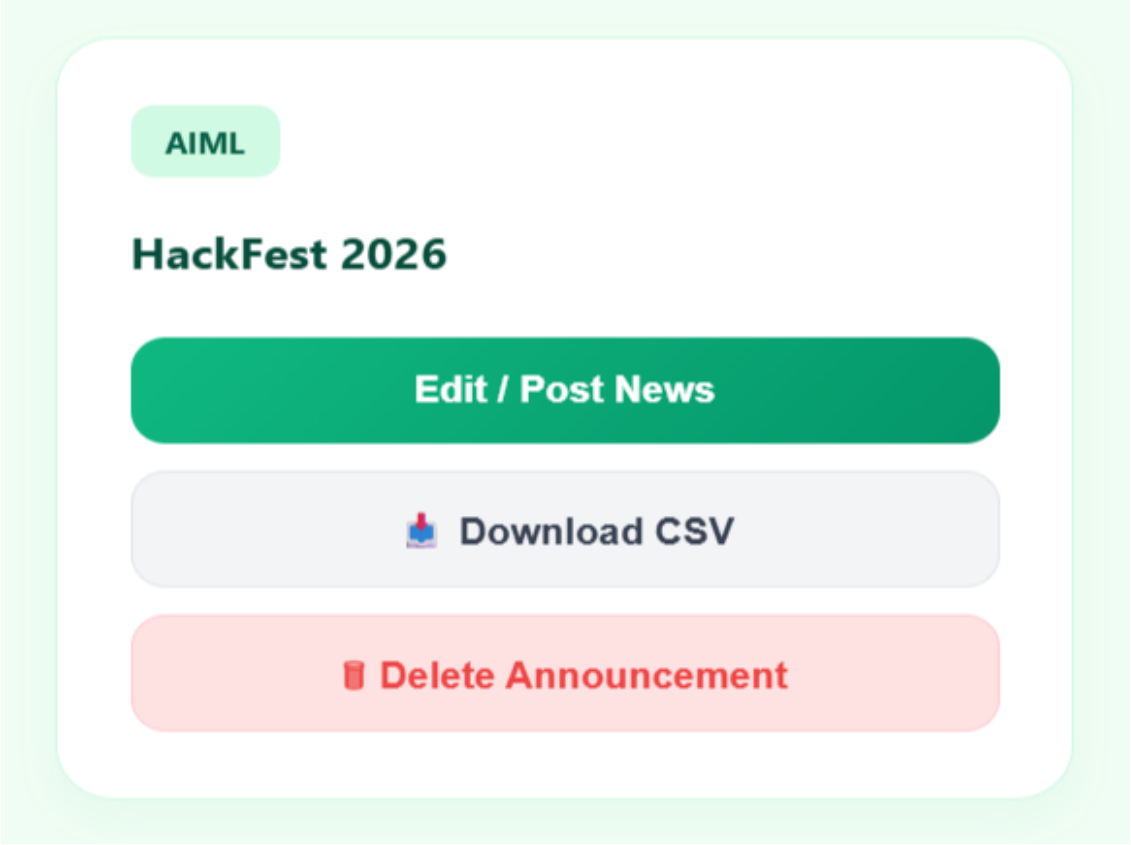

Figure 11:Download CSV

Chapter 8
Conclusion and Future Work

8.1. Conclusion

The **Centralized Hackathon Management Portal** has been successfully designed and implemented to address the challenges associated with traditional hackathon management systems. The project provides a unified platform that simplifies the process of organizing and participating in hackathons within a college environment.

The system effectively eliminates the need for multiple tools and manual processes by integrating all functionalities into a single interface. Students can easily access information about upcoming and ongoing hackathons, view detailed descriptions, and obtain problem statements during active events. This improves accessibility and ensures that all participants receive timely and accurate information.

One of the key achievements of the system is the implementation of **automated round progression**, which reduces manual effort and enhances transparency in the selection process. The system also ensures efficient data management by storing user and event information in a structured database, allowing easy retrieval and tracking.

Overall, the proposed system improves efficiency, reduces errors, enhances user experience, and provides a scalable solution for hackathon management. It successfully fulfils the objectives defined at the beginning of the project and demonstrates the practical application of web development technologies in solving real-world problems.

8.2 Future Scope

The Centralized Hackathon Management Portal can be further enhanced by adding advanced features to improve functionality and user experience. Future improvements may include real-time notifications and email alerts for announcements and deadlines, integration with authentication systems like Google login, and a more detailed analytics dashboard for tracking participation and performance.

The platform can also be expanded to support multiple colleges or organizations, making it scalable for larger audiences. Additional features such as project submission tracking, leaderboard systems, certificate generation, and judging modules can make the system more comprehensive.

Furthermore, implementing mobile application support, AI-based recommendations for hackathons, and enhanced security measures can significantly improve accessibility, personalization, and reliability of the platform to ensure transparency and fairness in the selection and evaluation process

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